POLS 4728 design language

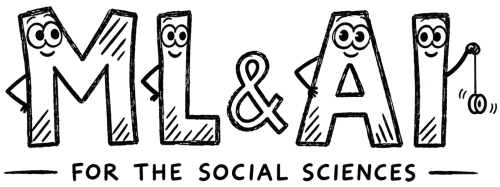
Scholarly notes with a playful visual identity.

A welcoming identity

Primary / outlined characters



Playful / the yo-yo variant



Use character artwork on covers and part openers, with occasional chapter illustrations that clarify a statistical idea. Develop concepts through explanation, mathematical reasoning, evidence, and code in connected paragraphs. On the notes cover, place the complete original yo-yo logo high on the page. Its embedded subtitle supplies the visible title; do not repeat it in a separate typeset heading. Retain the full course title in the PDF metadata.

Document accents and plotting colors

Blue and rust anchor the document. Plots use an eight-color cycle with matching colors for related marks, plus labels or patterns that survive grayscale printing.

Blue Rust Teal Purple

Ochre Sky Rose Gray

A familiar mathematical voice

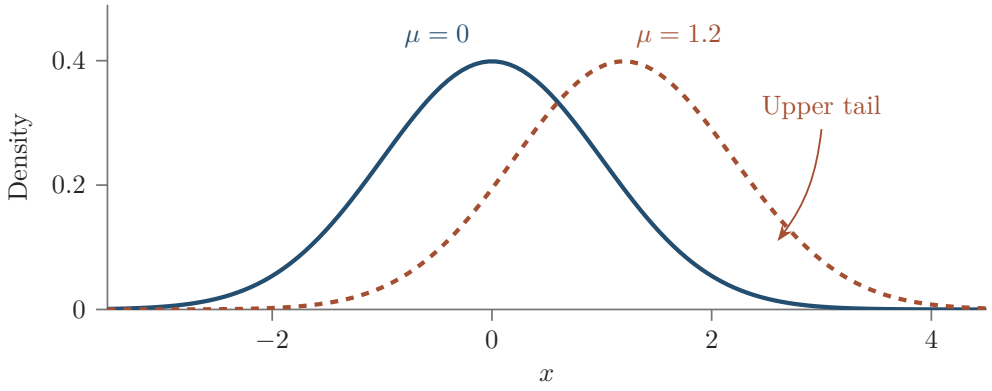
Latin Modern text and mathematics retain the character of a TeX textbook. Use clear hierarchy, open space, and readable labels at the final printed size.

$$\widehat{y}_i = f_\theta(x_i), \qquad \text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2.$$

Print defaults: white backgrounds, 11 pt body text, 10 pt plot labels, thin rules, and no decorative page textures.

Exposition, figures, and code

Open textbook, with a useful annotation



Two unit-variance normal distributions. Moving the mean from 0 to 1.2 shifts probability to the right. Labels and line patterns identify both curves even when the page is printed in grayscale.

ROC and precision–recall plots keep all four borders around $[0, 1]^2$, overriding the open-axis default. Label zooms as detail views.

Develop explanations in connected prose, preserving assumptions and derivations. Keep useful lists for vocabulary, assumptions, examples, comparisons, and procedures. Avoid fragmented paragraphs, forced line breaks, punchy lead-ins, decorative boxes, quirky pull quotes, story sidebars, and bold body emphasis. Use `\emph{}` for new vocabulary, such as *training data*, or necessary stress. Keep later mentions and routine lead-ins in regular type.

Code students can read and reuse

Use `listings` for selectable, monospaced text on white. Aim for at least 10 pt at printed size. Keep lines short, preserve indentation, and use line numbers only when the explanation needs them.

This example from the Python notes squares each prediction error and adds the results. The output is 14.

```
errors = [1, -2, 3]
total = 0
for error in errors:
    total += error ** 2
print(total)  # 14
```

Check readability in grayscale and paste a sample into an editor: indentation, punctuation, and line breaks must survive. Bold keywords and italic comments support the colors. Shorten long examples before reducing the type size.

Illustration as explanation

A reference from *Statistics*

Freedman, Pisani, and Purves's *Statistics* (4th ed., W. W. Norton, 2007) is a reference for the relationship between exposition and illustration: direct prose, concrete examples, economical diagrams, and occasional visual wit. Use sparse black line drawings on white backgrounds, with simple compositions and little shading. Within chapters, an illustration should clarify a mechanism, reveal a misconception, or make a comparison memorable. Humor should arise from the statistical idea.

Conceptual drawings and quantitative figures

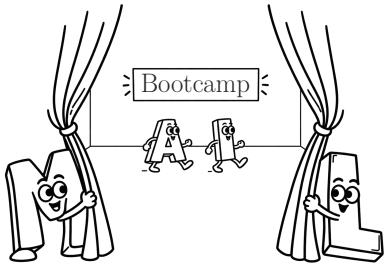
Hand-drawn character is welcome in conceptual illustrations; quantitative figures must retain precise geometry, readable labels, and faithful scales. Preserve the course's established typography, palette, and characters, and develop original illustrations suited to its subject matter. Conceptual drawings must not imply numerical relationships that the explanation does not establish.

A place within the argument

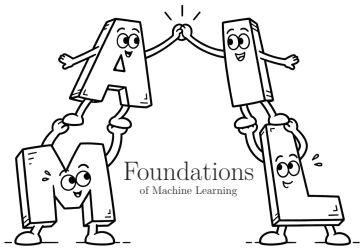
Place each illustration near the passage it explains. Its caption and the surrounding prose should state the learning point; essential labels and mathematical notation remain real text. Use illustrations selectively when they advance the argument. Keep the existing rules for scholarly prose, ordinary headings, and restrained formatting.

Use a clear silhouette and legible details at the final printed size. Register new character artwork in `design/assets.json` with a description, hash, and intended use, and keep image-generation prompts outside version control. Review the result in grayscale and describe meaningful visual content in alternative text where the format supports it.

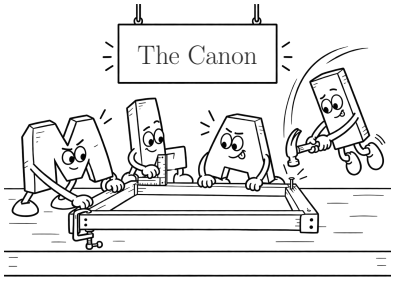
A welcoming start to each part.



M and L open the curtains; A and I march farther back on the stage.



M and L support A and I, whose joined hands complete an arch.



The letters build together, with I airborne during a hammer swing.

Typeset one title inside the unlettered scene using regular-weight Latin Modern. Keep it selectable and preserve the contents entry and PDF bookmark. Use a small part number above the composition and a quiet folio. Balance the whole group slightly above the page's center, at about 90% of the text width.

Preserve recognizable characters, sparse black outlines, and the full scene. Keep draft notices in introductory prose. Keep the original artwork and register derivatives in `design/assets.json`. All three openers use the same composition helper in the guide and notes. *The Canon* begins with ridge regression after the foundational shrinkage and collinearity discussion.

Python source beside each listing

Use the *Matplotlib for Storytellers* pattern: a readable `.py` filename at the listing's upper right opens its GitHub source page. Five active listings use maintained local source files and 10 pt type. Their public copies live in `pols4728`, while the semester repository remains private.

The shared helper prints the local file and links to a matching public commit. When an example changes, publish its updated copy, update the shared revision, and rebuild the notes. Check that each link works without signing in and that the downloaded file matches the printed source.

Readable in more than one way

Pair color with words, shapes, and line patterns. Keep essential information in text and check pages at actual size and in grayscale.

Accessibility status. This system improves visual readability and authoring practice. The current PDFs are untagged; screen-reader structure, accessible mathematics, and embedded alternative text need a separate build migration. The complete rules and migration register are in `design/README.md`.

Scrollytelling and interactive explainers

Scrolling carries the explanation

Every course visual explainer must be a working scrollytelling experience unless the user explicitly requests another format. Keep the main graph, plot, diagram, or equation pinned while scroll position constructs or transforms it and brief text floats in and out. Storyboards and prototypes must demonstrate that motion. A long illustrated document, entrance fades alone, or timer-driven playback is insufficient. Use substantial visual construction and transformation to carry the argument.

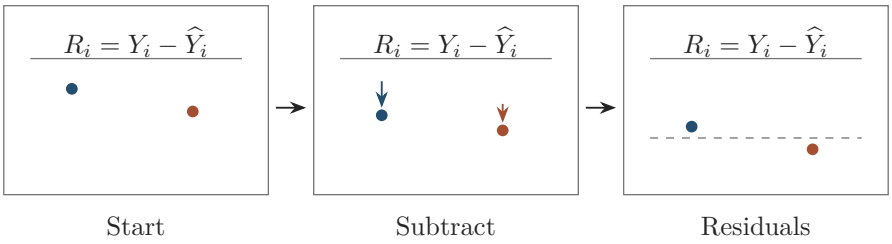
Title slides and opening scenes

Begin a scrollytelling lesson with a title scene. Explainer openings and illustrated lesson title slides usually pair a real-text title with torso-less figures acting out a metaphor: adjusting a picture frame, hammering out a sine wave, or another simple action that captures the central idea. Use one spare monochrome scene, few props, and generous white space. Preserve the full scene and a selectable title separate from the artwork. Let scrolling control any opening animation. Retain existing deck, course-cover, and part-opening conventions.

Torso-less figures and spare text

Human characters use one expressive round face-and-body shape with arms and legs attached directly, no neck or separate torso. Match the registered artwork: sparse black outlines, white interiors, simple hands and feet, and little shading. Let gestures serve the explanation; keep quantitative geometry precise and essential labels and mathematics as real text.

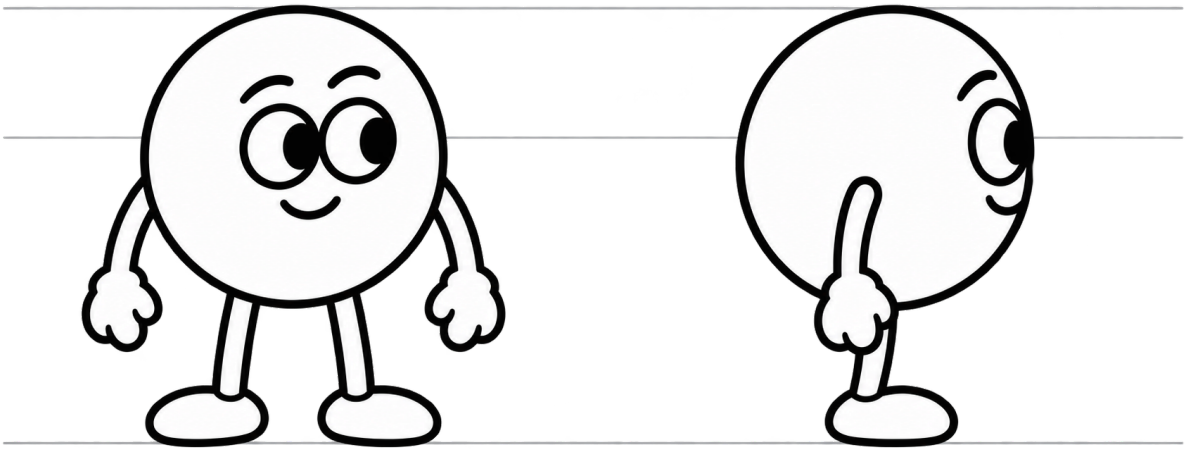
Let the visual dominate a white field with generous open space. Give each scroll beat one explanatory task. Use short headings and concise captions or explanations where needed. Reveal longer arguments in successive states while preserving their assumptions, qualifications, and conclusions. Let diagrams and equations do most of the explanatory work.



A pinned equation anchors a changing diagram. These frames specify a conceptual progression, not a numerical dataset: the same marks move as scrolling subtracts predictions. The accompanying text explains each step.

Torso-less character reference

The same figure is shown head-on and in profile, following the SHAP character family. Use this specimen as the anatomy reference for new human characters.



Head-on Profile

One round shape serves as both face and body. Arms attach directly to its sides and legs to its lower edge; there is no neck or separate torso. Keep the same scale, proportions, and rounded hands and feet across views. The faint lineup rules make the alignment visible.

Use sparse black outlines and white interiors on a white or transparent field. Keep view labels, titles, and mathematical explanations as real text outside the artwork. Expressions and gestures may change while the anatomy stays recognizable.

Scroll controls and review

Scrolling is the complete route

Use scrolling as the default and preferably only storytelling input. Core teaching points must need no toggle, slider, play button, click, or hover. Tie meaningful intermediate states to scroll position as objects, observations, parameters, comparisons, and equations change. Reserve optional controls for purposeful exploration after the relevant sequence, only when scrolling cannot supply the learning benefit. Preserve native scrolling and keyboard access.

Pinned graphs and floating text

Keep the main graph or plot pinned while brief text floats in and out with scrolling. Apply this to diagrams and equations too. Coordinate each arrival, readable hold, and departure with the visual change; keep marks, axes, labels, and equations clear. The plot changes internally within its fixed stage. Reverse scrolling restores earlier visuals and text.

Specify each scene

Record the learning point, pinned visual, scroll interval, transformation, accompanying text, and completed state. Meaningful intermediate states follow scroll position and reverse with backward scrolling. Preserve observation identities, scales, and color roles across related scenes. Hold each state long enough to read. Standalone lessons must not name the source assignment, quiz, or question number.

Keep the experience usable

Retain a compact pinned graph, diagram, or equation on narrow screens, with room for text to enter, remain readable, and leave. Provide keyboard navigation. With reduced motion, show complete, legible states at logical scroll thresholds without continuous movement; preserve the full argument and results.

Verify actual scrolling

Scroll the actual page forward and backward on desktop and narrow mobile layouts. Check plot pinning, text arrivals and departures, intermediate geometry, reverse progress, keyboard access, reduced motion, labels, and final values. Confirm access by scrolling to every teaching point, spare prose, torso-less characters, and a clear title metaphor. Final-state screenshots and unit tests alone are insufficient. See `design/README.md` for complete authoring and review requirements.